

Andrew Harnor Curriculum Vitae

Security Clearances Held: MOD SC; HO SC; lapsed NPPV3, **Nationality:** British, **Resident:** UK (Bucks).

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Summary

Andrew is a British engineer who achieved Chartered Engineer status. He has extensive experience in programming, Linux implementation, scripting, and customer support. Skilled in multiple programming languages, system administration, and practical technical delivery, he has worked extensively in Unix/Linux environments and has a strong background in CAD/CAM systems. He brings a broad range of technical skills, project success, and career achievements across several industries.

He is a practical engineer with sound judgement and a habit of quietly improving systems and workflows through useful, workable solutions. He also has a track record of making small, pragmatic changes in customer environments that significantly improve software usefulness and productivity.

He is now looking to apply that combination in useful modern technical work such as practical AI implementation, CAD/CAE/PDM manufacturing systems, infrastructure-as-code/Ansible, or CI/CD-related delivery.

Extensive technical skills: Proficient in Linux administration, Red Hat, Ansible, Docker, scripting languages (Perl, Bash, Python), and broad skills in programming languages (e.g. Fortran, C, C++, Java), plus experience in SQL and Oracle databases.

Personal Qualities: Thoughtful, practical, dependable, quietly determined, quietly effective, with calm judgement, a measured approach to substantial technical work, and a collaborative, non-ego-driven style.

Personal Statement

I am an analytical graduate engineer with a cautious but determined approach, preferring to think things through before acting. I prefer to work towards defined project targets, but I can switch focus to day-to-day issues when they take priority. Starting as a developer in Fortran and C, I have since worked across the full project life cycle, including development, scripting, Linux/system administration, product support, on-site customer work, and technical implementation.

I have regularly worked between end users and developers to co-ordinate, test, and support changes as they move towards production. A particular strength is using development knowledge with customers and users to identify pragmatic changes, fixes, or refinements that turn nearly-ready products into something useful in practice.

I can produce detailed technical documentation and planning material, but I am equally comfortable working from a supplied brief, informal discussion, or a loosely defined requirement where the objective is clear. I also identify and initiate useful technical work independently, provided I am confident the effort will produce practical results. I can independently deliver a project over a six- to twelve-month timescale, including cases where the work fits into a larger programme.

My fundamental technical strength is excellent knowledge of the Linux command line and Unix utilities. My current strengths lie in Linux/system implementation and practical technical delivery, supported by deep software foundations, issue ownership, and substantial manufacturing engineering systems experience.

Skills and Experience Matrix by Theme

Theme 1 – Linux Platform and Technical Implementation

1986–Present	Unix System Support	Customer projects, help desk, on-site support, migration, system improvement, Ansible build support
1990–Present	Unix/Linux Administration	Installation, maintenance, scripting, system design, practical patching, problem resolution, BAU support, Linux kernel parameter config, SSH keys/PKI certificates, Linux hardening, CIS compliance
1997–Present	Unix Server Administration	Platforms: Red Hat, AIX, Solaris, Ultrix, IRIX; storage disk array, LV, SELinux, High Availability/HA-CMP, Veritas VCS
1997–Present	Clusters and Failover	Veritas VCS, Oracle Grid/RAC, LV mirroring, HA/failover support
2016–2026	General Administration	dnf/yum, apt, NTP, DNS, AD, Firewall, TCP/IP, UDP, etc.
2001–Present	Monitoring and Operational Support	Nagios, SCOM, BMC Patrol, SNMP, operational monitoring, alert-driven escalation, incident work, change-window support
2016–2026	Virtualisation	VMware, ESXi, NSX, HCX, Hyper-V
2016–2023	Modern Tools	Ansible, Ansible Tower, playbooks, Satellite, YAML, Git

Theme 2 – Practical Use of Programming Languages

1982–1986	Fortran	Designed, developed, and implemented Medusa and CAD/CAM applications.
1986–1998	C / Fortran77	Development, support, debugging, and testing of very large monolithic C/CADDS C/Fortran77 codebase
1998–2020	Perl	Perl applications, Perl modules, object-oriented Perl, Perl web development
1999–Present	Java / C++ Awareness	Learned Java and C++, basic language usage. Created applets, feasibility testing. WAR-file build
1982–Present	Skilled Code Refinement	Careful investigation, safe debugging, careful judgement, technical escalation
1983–Present	Mix of Analytical and Strategic Problem Solving	Strategic and instinctive thinking, wider-situation awareness, effective technical go-between
1986–2023	Source Control	Proficient in Git, GitHub, Subversion, and ClearCase

Theme 3 – Data and Structured Information

2000–2016	Oracle/MySQL/PostgreSQL	Relational database installation, schema design, admin, SQL queries, PL/SQL.
2003–2010	Oracle/SQL Server	PDM support, Perl:DBI, fairly complex SQL, triggers, and stored procedures
1995–Present	Ticketing and Information Flow	Agile, Jira, call handling tools, Remedy, ITIL, ticketing data, information flow
2011–Present	Other data formats	MatrixOne MQL, WAR files, XML, JSON
2016–Present	Other database types	Basic knowledge of MongoDB, NoSQL, and related non-SQL technologies

Theme 4 – Scripting, Automation, and Practical Tooling

1989–Present	Shell Script	Small/large application scripts, various shells (Bash, Korn, and csh)
1998–2020	Perl	Object-oriented Perl, Perl applications, Perl modules, and Perl web development
2009–2023	Tools/Utilities	XML, YAML, ad hoc Ansible, modern Bash constructs, tmux, PowerCLI with VMware, Tomcat, LAMP

Theme 5 – Customer Collaboration and Technical Bridging

1988–2022	Technical Product Refinement/Customisation	Working between developers and customers to implement changes, support delivery, and clarify technical issues
1994–2000	Front-line Product Support	Telephone support, data translation issues, customisation, practical response to customer requirements

Theme 6 – CAD/CAM/CAE Enterprise Implementation

1982–1984	Medusa DARS programming	Developed bespoke applications for use with Medusa Design.
1983–2006	CAD numerical methods	Support/develop 3D algorithms (e.g., finite element and surface design)
1984–2000	CADD55 development	Develop, support, and debug Computervision 3D CAD/CAM software
1997–2006	CATIA V4 / PTC ProE	Installation, administration, user support, implementation within engineering-system environments
1998–2018	EPD.Connect, PDM	Product installation, PDM/EPD.Connect vault administration, workflow
1999–2004	Composite materials	Uses and benefits, bugfix programme, product definition, component development
2011–2018	Enovia/CATIA V5	Installation, upgrade, and configuration of CATIA V5/Enovia PDM; Enovia customisation
1987–2006	Data exchange	IGES, STEP, VDA-FS, Theorem Solutions Converter, PDES

Employment History

Jul 2023 – May 2026	Linux Migrations	DXC , home based/Buckinghamshire (Permanent)
Jan 2021 – Jul 2023	Linux Support Role TSS12	Fujitsu Defence & Security , Bracknell/Home (Permanent)
Apr 2019 – Feb 2020	Red Hat Second Line	Vodafone/IBM Joint Venture , Swindon (Contract)
Sep 2018	Red Hat BAU Engineer	Arqiva Smart Metering , Winchester (Contract)
Nov 2016 – Jun 2018	Linux/Unix Admin / Scripting	Airbus Defence & Space , Stevenage (Contract)
May 2015 – Jun 2016	Perl/Open Source/Unix - Docker	IBM (Hursley) , Winchester (Contract)
Nov 2013 – May 2015	Unix/Ubuntu/Java - Perl/TCP/IP	Various Small Projects (Contract)
Nov 2011 – Nov 2013	Perl/Unix/Shell - Enovia Support	Jaguar Land Rover/IBM , Coventry (Contract)
Apr 2011 – Jun 2011	C/Unix/Perl - Network Development	AceAxis Networks , Aylesbury (Contract)
Jul 2010 – Nov 2010	Perl Web Development	GE Energy , Livingston (Contract)
Feb 2010 – May 2010	Perl/Unix - Script Development	HMRC/CapGemini , London (Contract)
Jan 2009 – Jun 2009	Project to fix Non-Compliant XML	Wiley/Blackwell (Scientific Documents) , Oxford (Contract)
Apr 2007 – Jul 2008	Migration + Test with Perl Scripting	BBC Monitoring , Reading (Contract)
Jan 2007 – Mar 2007	Perl Development	BBC Wales , Cardiff/home (Contract)
Oct 2006 – Nov 2006	Perl Web Development	VisionOSS , Reading UK (Contract)
Aug 2006 – Sep 2006	Perl Script Development	NetApp/Company-i , work from home (Contract)
Nov 2002 – Apr 2006	Perl/Oracle - Scripts	IBM/Airbus UK , Bristol (Contract)
Jan 2002 – May 2002	Unix Server Support/Help Desk	Nortel Networks/CSC , Friedrichshafen (Contract)
Nov 2000 – Jul 2001	Concept Design of Monitoring Tool	Otto Versand , Hamburg (Contract)
Aug 2000 – Sep 2000	Unix Shell Script Development	Nortel Networks/EinsteiNet ASP , Hamburg (Contract)
Jan 1998 – Jul 2000	On-Site Unix & CAD/CAM	PTC-Computervision/EADS/Airbus (Contract)

Aug 1997 – Jan 1998	Support Unix Client Server Administration	Debis System House/Dasa , Munich (Contract)
Jan 1994 – Aug 1997	Senior Helpdesk Support Analyst	Computervision UK Services , UK and Germany
Apr 1986 – Jan 1994	Unix SysAdmin/Snr Software Engineer	Computervision , International Software Support
Jun 1985 – Apr 1986	Software Engineer	PA Management Consultants
Jun 1984 – Jun 1985	Software Engineer	Computervision , European Development Group
Aug 1982 – Jun 1984	Technical Programmer	Dowty Mining Equipment

Reversed Chronological List of Recent Roles

(with thematic highlights and supporting evidence below)

2023–26 – DXC – Worked on the migration of large Government applications (RHEL, SUSE, Windows) into a secure hybrid cloud platform based on VMware, with some physical infrastructure. On my own initiative, I developed scripts because I could see the benefit to both the migration process and the associated tooling conversions, including NetBackup, Dynatrace, and antivirus tooling. I carried out my share of the migrations to a high standard using Carbonite, HCX, and LV mirroring techniques, and helped the team handle storage-migration issues. In addition, I developed PowerShell scripts for basic VMware actions (snapshots, tagging, affinity rules), and gained practical VMware skills together with an understanding of VMware NSX concepts including Endpoint, Tunnelling, Management Plane, Data Plane, permissions, and North/South traffic.

Recent manager feedback

“One of Andrew’s standout strengths is his scripting and automation capability. The tools and scripts he created have streamlined tasks, reduced manual effort, and significantly improved the efficiency of key migration activities.” —

Migration Programme Manager

“He reliably delivers on commitments, doing what he says he will do and always demonstrates a high standard of quality and accountability.” — Direct Project Manager

2021–23 – Fujitsu Defence & Security – Worked in a security-controlled Linux support and implementation environment for a large Java-based biometrics application, using Satellite, Ansible Tower, Git/GitHub, YAML playbooks, and Bash tooling across the SIT-to-PROD environment range. The role combined Linux/platform administration, controlled change and deployment activity, practical automation, and cross-team delivery in a complex Fujitsu/Home Office/IDEMIA setting. See Themes 1, 2, 4, and 5 for further detail.

2019–20 – Vodafone/IBM Joint Venture – Provided patching support on large Government/Defence platforms in a security-cleared, ITIL-controlled operational environment, supporting Linux patching, controlled change, incident response, and associated VMware activity. Work was tightly controlled within defined patch windows, with significant time pressure and testing to ensure systems were returned to service correctly. On my own initiative, I improved the patching process through practical scripting and automation, enabling patching throughput to be significantly increased. See Themes 1 and 4 for further detail.

Sep 2018 – Arqiva Smart Metering – Carried out BAU Linux support in a resilient, high-availability environment, covering patching, network configuration, failover support, hardening, and compliance-related work. See Theme 1 for further detail.

2016–18 – Airbus Defence & Space – Delivered Linux/Unix administration, Ansible-based build, Red Hat Satellite, monitoring, patching, SELinux, and migration work across Red Hat/CentOS platforms, combining day-to-day operational support with practical technical improvement. See Themes 1 and 4 for further detail.

The role also included broader support across Oracle and Tibco environments, plus database migration and sizing work, migration of HP-UX packages between disk arrays on a cluster using Serviceguard, and related storage work. I improved Nagios monitoring and wrote Bash/Perl scripts for data feeds and API/CLI access to Red Hat Satellite.

2015–16 – IBM (Hursley) – Worked on Docker-based porting of open-source tools and Hadoop components for Linux, combining practical container build work with hands-on problem-solving in the core Hadoop module. Also contributed supporting database and socket-programming work. See Theme 2 for further detail.

Scanned open and closed-source applications for right-to-use, copyright, and IP issues. Modified a Java application to scan packaged source code and record thumbnail image data in a spreadsheet, using ImageMagick and Apache packages including Swing, POI, Batik, Log4j, and xml-apis.

2011–13 – Jaguar Land Rover – Contributed to a large project to customise JLR's enterprise-wide iPLM platform for CAD/CAM/CAE manufacturing processes. See Themes 3 and 6 for further detail.

Thematic Capability Highlights with Supporting Evidence

Theme 1 – Linux Platform and Technical Implementation Highlights

1990–94 – Computervision – Administered a network of CAD workstations used to develop and bug fix CADDs software, and supported developers and fixers in many ways. “Left the environment a lot better than you found it” — Chris Boutal, Manager.

1997 – Debris (Munich), Germany – Worked with HA-CMP, logical volumes, AIX, and Solaris.

2016–18 – Airbus Defence & Space – Gained a solid understanding through use of an Ansible-based build system using JSON over HTTPS. Also carried out database migration and sizing work, participated in storage disk-array upgrade work, modified **SELinux** policies to allow monitoring within a secure environment, improved **Nagios** monitoring, and reviewed monitoring trends to improve performance. Played a major part in turning “a Nagios jungle of ignored alerts” into “a fairly well-tuned system, useful for monitoring the systems and sending alerts to the right people” — Amarjit Ball.

2018 – Arqiva Smart Metering – Carried out BAU Linux support including patching, network configuration, VMware snapshotting, Veritas VCS failover, Red Hat Satellite registration, Linux kernel parameter tuning (MTU), Splunk upgrades, SNMP configuration, Linux hardening, SSH keys/PKI certificates, and CIS compliance work.

2019–20 – Vodafone/IBM Joint Venture – Carried out Linux/Unix patching and associated incident work on large Government/Defence applications across environments such as QA, DR, and PROD, under ITIL change control. This required cluster separation and reconnection, Oracle Grid/RAC restart under DBA direction, JBoss stop/start, and minor change, reconfiguration, and update work (e.g. NetBackup, VMTools, and antivirus), together with associated VMware activities (intermediate level) and a small amount of Solaris 10 patching. A significant part of the role was ensuring that systems were returned to service correctly at the end of tightly controlled change windows.

2021–23 – Fujitsu Defence & Security – Worked in a **security-controlled Linux support environment**, using **Satellite, Ansible Tower**, playbooks, and related tooling. Practical Linux administration under **ITIL** change control required attention to compliance as changes were promoted through standard environments from **SIT to PROD**. Also used **Hyper-V**, with some exposure to storage and Linux/Windows interoperability via access control, **realm**, and **Samba**.

Theme 2 – Practical Use of Programming Languages Highlights

1982-84 – Dowty Mining – Designed and enhanced technical Fortran utilities, particularly EASOF (a finite-element solution for beam deflections on a foundation) and a graphical BOM/CAD data tool.

My natural approach is to step back from the immediate task, understand the wider workflow, and improve the overall system rather than just the immediate problem.

At Dowty I soon realised that the existing programs were fragmented spaghetti code, with identical functions repeated throughout. I built shared libraries for User Interface, CAD Access, and Data Handling, and although I was cautioned that this would be a risky change to a long-established codebase, its value became clear as the common libraries were adopted across the programs. This proved key to unlocking the productivity needed to deliver the BOM tool within a short timescale.

1982-Present – Skilled Code Refinement – I am skilled at delving into complex source code, including unfamiliar languages or products, to get sufficiently to grips with the moving parts and quickly decide the best approach for practical improvement.

In practice, this has often meant working as a technical generalist in specialist areas where I am not expected to be the specialist. I apply tenacity, care, consideration, and practical judgement to decide whether a safe, useful fix is possible, or whether deeper algorithmic, product, or language knowledge is needed.

Where I judge that a safe, useful fix is possible without drawing on specialist knowledge, I can do so; where the issue needs deeper expertise, I analyse all the available evidence and present clear technical findings so a specialist can act.

I have applied these skills repeatedly across different products and problem domains throughout my career, including EASOF, CAD graphics generation, NC methods, and 3-dimensional tensor mathematics.

1983-Present – Mix of Analytical and Strategic Problem Solving –

I am generally effective in support and customer-facing problem-solving roles: I quickly get to the nub of a problem, understand what the customer is really trying to achieve, and use development knowledge to move things forward in a way that is both technically sound and genuinely useful.

I evaluate issues at several levels, combining strategic thinking with practical instinct. I absorb clues and evidence from the wider situation, customer context, product behaviour, and source code to form a judgement about what is really needed.

This judgement is not guesswork; it comes from hard thinking, great care, and processing the whole situation until the real issue becomes clear.

Because I can absorb both end-user and product-specialist viewpoints, I am an effective go-between where reaching a solution requires good collaboration between the customer and product specialist.

1986-90 – Computervision – Fixed critical defects in CAD/CAM software. Consistently each year, achieved the **second-highest fix rate** among more than **50 bug fixers** based in the USA and UK. (I focused mainly on IGES, Surface Design, and 3/5 Axis NC Software.)

1987-88 – Computervision – Received an award for enhancing the CGOS Text Editor to allow basic regex searches.

2011 – AceAxis Networks – Network Programming in C and Perl – Knowledge of Mobile Phone Protocol UMTS/(W-CDMA).

2015-16 – IBM (Hursley) – Used Docker as a development environment to port open-source tools including gcc, MySQL, Zookeeper, MongoDB, and Joomla to IBM 'Linux on Z'. Created Dockerfiles, entrypoint scripts, and network/port configurations, built orchestrated container sets to demonstrate working application examples, and published Docker recipes in Markdown. Worked within a team of 10 building Hadoop component containers. The Dockerfiles were submitted to Jenkins to demonstrate a working Hadoop CI/CD environment. Took responsibility for porting the core Hadoop module, resolving Big Endian and build issues, developing workarounds for assembly-code segments, and contributing code checks for other components. Also configured a MySQL database for web-data collection and developed TCP/IP and UDP client/server socket programs in C/C++, Java, and Perl.

2021-23 – Fujitsu Defence & Security – Developed **YAML** Ansible playbooks, with my changes managed through **Git/GitHub** and uploaded to **Ansible Tower** and **Satellite**. The wider Biometrics project involved implementation of a large Java-based application. I developed the solid working knowledge of **Java**, **XML**, and **JSON** required to configure the system and investigate issues, together with an understanding of open-source components including RabbitMQ, Cassandra, Zookeeper, Cluster, and **AMQP**. We worked in cross-functional **Agile** teams (Linux, Windows, Storage, Database) using **Jira** and formal **ITIL** change control, with changes promoted through standard environments from **SIT to PROD**.

Theme 3 – Data and Structured Information Highlights

2000 – Otto Versand, Hamburg – Oracle Database Design and Administration training (in German)

2003 – Airbus (Bristol/Filton) – Wrote SQL scripts, including a vault-independent attribute lister and a generalised Oracle Perl:DBI EPD-Connect vault scanning tool. Recognised as useful by Phil Holder (Manager) and Julian Brown (Consultant).

2010 – GE Energy – Developed complex SQL searches to extract the change data from ClearCase, creating a searchable archive of product change history.

2013 – Jaguar Land Rover/IBM – Assisted with the introduction and customisation of the Enovia iPLM CAE environment, including installation of Enovia components, building Tomcat WAR files, minor Java changes, and work with MQL — MatrixOne's SQL-like query language.

2001-2016 – Various – Basic database schema design, table creation, database administration, some MongoDB and NoSQL, across several projects.

Theme 4 – Scripting, Automation, and Practical Tooling Highlights

1989 – Computervision – Designed and created a shell utility to build and distribute CADDs patch executables from source code. It checked each fix for conflicts with other patches, maintained records of all patch distributions, tracked patch upgrade histories across CADDs revisions, and marked fixes integrated into the core product. I also added searchable patch-database functions so existing fixes could be found quickly and supplied to customers reporting similar issues. The utility was used daily across Europe for 10 years with minimal changes, and I was told, "**You did a lovely job of that**" — Colin Balmforth, Support Manager.

2005 – Airbus/IBM (Bristol/Filton) – Created a Perl Secure FTP (Zamiz) script to mirror/replicate CAD parts around Airbus vaults.

2006 – NetApp – Developed a Perl utility to take NetApp snapshots for a confidential project in Cheltenham. The performance appraisal states: "Delivery Exceeded Expectations ... was a pleasure to work with ... Unix/Perl knowledge was key to the project success."

2007 – BBC Monitoring – Created shell scripts to support the migration of outgoing BBC News feeds during a software/MySQL upgrade. The project ran under ITIL conditions. "**Your cautious can/will do attitude has turned a potentially serious customer-impacting situation into a massive success as far as getting this system into production. I can truthfully say it would not have happened anywhere near as quickly and safely without your efforts.**" — Clive Woodley.

2009 – Wiley/Blackwell (Scientific Documents) – Created a Perl utility to fix 80K non-compliant XML scientific documents and bring them in line with the Wiley DTD. Used various Perl XML libraries for much of the work.

2010 – GE Energy – Developed a Perl utility to extract details of software changes (change data) from the **ClearCase** source control system to create a database of release bulletins for distribution to end users. This required complex SQL searches to extract the change data, and used ClearCase libraries, Perl CGI, and web servers to make the bulletins searchable and distribute them using Perl Template. The project was budgeted at 75 days and completed in 72.

2013 – Jaguar Land Rover/IBM – Developed a Perl-based multi-page web interface utility to maintain P&O user, project access, and Enovia licence allocations for 200 users across 20+ sandbox environments. Created Perl sockets to distribute and update the sandboxes from a single console.

2016–18 – Airbus Defence & Space – Developed Bash and Perl scripts for data feeds to space satellites, network connections (**lftp, sftp, rsync, ftp, ssh, chroot**), and scripts to access Red Hat Satellite via its API and Hammer CLI. Wrote Nagios plugins and related tooling (Bash/Perl/Python and a little C/C++) to improve Nagios monitoring and support Oracle/MySQL monitoring.

2019–20 – Vodafone/IBM Joint Venture – On my own initiative, developed scripts (PowerCLI, Bash, Perl, PHP, Oracle DBI) to automate patching tasks and halve patching time. **“What you have developed is quite ingenious and has produced a discernible patching process that ought to be adopted widely across the team.”** — Nick Witbooi.

2021–23 – Fujitsu Defence & Security – Initiated and continuously developed an environment-independent library of ad hoc **Ansible** commands and **Bash** tooling to support repeatable operational tasks across the **SIT-to-PROD** environment range. The tooling coped automatically with environment differences, such as varying numbers of message-handling or database components. The same library command could perform the same task, such as starting or stopping a service type, consistently in differing environments. In parallel, I improved my **Bash** scripting by using modern Bash constructs to create scripts and processes that could communicate with each other, wait on dependent activity, and tidy up cleanly on a Linux **SIGNAL** or if something failed.

The library also grew into a catalogue of environment-independent fixes and investigation queries, updated as new problems were encountered and then reused to support fast responses to regular **SCOM** and **McAfee** alerts. I used **tmux** to manage, monitor, and switch between multiple live sessions while running support tasks, which streamlined the support role when several actions needed to be watched or co-ordinated at once.

2023–26 – DXC – Developed scripts to support migration and tooling conversions, including backups, Dynatrace, and antivirus. See Recent Roles above for fuller context.

Theme 5 – Customer Collaboration and Technical Bridging Highlights

The care, consideration, and practical judgement described in Theme 2's practical programming work give me practical effectiveness in customer-facing technical work, product implementation, and support situations where outcomes matter. I have worked in real operational environments, under pressure, with customers, users, delivery expectations, and real consequences.

1986–98 – Computervision – Developed expertise in CAD/CAM data exchange and became a European focal point, working with customers and Product Specialists in the USA, as well as European customers and local country offices.

1994–96 – Computervision – Call-Centre support for CAD/CAM software. Consistently achieved the second-highest call closure rate in the team.

1987-90 – Computervision – Led a very successful ongoing customer management project, working closely with Austin Rover to fix issues and deliver minor product enhancements. Also supported shorter-term projects at Ford, Rolls-Royce, and GE.

1998-2000 – PTC (at Airbus Hamburg) – Provided on-site support for CAD/CAM software. Re-worked reported issues and reduced the ongoing open bug count from 400+ in 1998 to about 25 in Sept 2000.

1999-2000 – PTC (at Airbus Hamburg) – Organised and chaired on-site Airbus/Computervision meetings to drive product improvement in Electrical Schematics. Later developed working knowledge of composite materials processes and helped drive a bug-fix and improvement programme that took the Computervision composites package from "unusable" to having practical production value.

2000 – PTC/Airbus – Germany/France/UK – Set up and participated in meetings between Airbus and PTC to specify a proposed development of a Composites package. (The Computervision Composite Product was only used in Germany)

2021-23 – Fujitsu Defence & Security/Home Office/IDEMIA – Worked within a complex multi-party delivery environment involving Fujitsu, the Home Office, and IDEMIA, where role-based access control (RBAC) tightly constrained participants. Routine support and deployment activity was therefore dependent on good co-operative relationships between the parties, with careful coordination and formal change control. Although my role was support-focused, it strengthened my ability to contribute effectively in situations requiring cross-team collaboration, controlled change, and practical technical judgement.

Theme 6 – CAD/CAM/CAE Enterprise Implementation Highlights

1982-2018 – Dowty/Computervision/PTC/MTU/Airbus/Daimler-Benz Aerospace/JLR/Airbus Defence & Space –

Extensive CAD/CAM/CAE and PDM experience gained in real engineering and manufacturing environments, applying graduate Mechanical Engineering skills, product design/drafting experience, software skills, and wider practical experience within engineering and manufacturing systems. This underlying understanding remains highly transferable within the field and to other real-world technical domains.

1982-1984 – Dowty Mining – Worked with mechanical product designers and engineers to develop practical engineering design software for pit props and related mining equipment.

1983-2006 – Post-processing NC CLData – Fixed bugs in NC numerical methods, and set up and tested CNC machine software at Filton.

1984-2000 – Computervision Development – Customer specific enhancements to CADDs, bug fixing in the CADDs product, and improved bug reporting.

1997-2006 – CADDs/Catia/ProE – Installed products, configured part storage, assisted designers with workflows, and supported specialised product commands.

1999-2004 – Airbus Composites – Gained understanding of the benefits of composite materials in aircraft design, composites software and related processes, and helped drive the definition of defects and enhancements to make the system productive, as well as discussions on future product direction. Also contributed to design work on a composite Druckerlotte (pressure plate), a large structural component fitted in the tail section of the Airbus A380.

2002-2006 – Airbus (Filton) – Work on Computervision's PDM product – Used Oracle SQL to correct PDM dataflow errors in response to vault requests (for example wrong revisions, attributes, or project assignments) and to produce ad-hoc reports.

2005–2013 – PDM/EPD.Connect/Windchill/Enovia – Database and data administration, workflows, policies, roles, and customisation.

2011–2013 – Jaguar Land Rover – Played a small part in a large project to customise Enovia, the core of JLR's enterprise-wide iPLM (Integrated Product Lifecycle Management) platform, connecting the product lifecycle from component procurement and vehicle design through to manufacturing and the dealership network. My specific role was to create a web console to configure People and Organisation (P&O) data for users, roles, policies, permissions, project assignments, and product licensing across multiple sandbox environments. I also carried out product installation and support work for Tomcat application servers and Oracle databases, together with minor customisation of some Enovia components using MQL (Matrix Query Language) and Java.

Education and Training

DXC Udemy (2026) – Various AI

Maidenhead Evening College (1989–1994)

Amersham Evening College (1992–1993)

Institution of Engineering Designers
(1993)

Cheltenham Evening College (1982–1983)

Brunel University (1978–82) - 4 Year
Course

Dowty Group Services (1980–1982)

School (1971–78)

Languages: English (Native), German (Fluent), French (Basic).

**Claude Code, Agentic AI, LLM Engineer, Agentic AI in
Production**

Studied German

Learned C++ Programming Language

Achieved Chartered Engineer status (CEng, MIED)

Obtained a Certified Diploma in Accounting and Finance (CdipAF)
BSc (Hons) Mechanical Engineering 2:1

Indentured Engineering Apprenticeship

2 GCE A Levels (Physics A, Mathematics B), **8 GCE O Levels**

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